

Sourav Selvaraj

souravselvaraj.github.io

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Research interests: legged robotics, whole-body control, and robot learning.

Education

- **Worcester Polytechnic Institute** Worcester, MA
Master of Science in Robotics Engineering Aug 2025 – May 2027 (expected)
 - **GPA:** 4.0/4.0
 - **Coursework:** Robot Control, Motion Planning, Optimal Control, Legged Robotics, Robot Learning
- **National Institute of Technology, Tiruchirappalli** Tiruchirappalli, India
B.Tech in Instrumentation and Control Engineering Dec 2021 – May 2025
 - **CGPA:** 8.2/10.0
 - **Coursework:** Control Systems, Robotics, Signal Processing, Machine Learning, Embedded Systems

Manuscripts

- [1] **S. Selvaraj** and M. Agheli, “SCRUB: Surface Contact Regulation Using the Body for Armless Quadruped Force–Motion Control.” Manuscript, 2026.
- [2] O. Sullivan*, **S. Selvaraj***, and M. Agheli, “GaitNet: Learning-Based Acyclic Footstep Planning for Dynamic Quadruped Locomotion.” Manuscript, 2026. *Equal contribution.

Research Experience

- **SCRUB: Armless Quadruped Body-Tool Sliding** ALMaS Research Group, WPI
Graduate Researcher; Advisor: Dr. Mahdi Agheli Aug 2025 – Present
 - Developed **SCRUB** for blind force-regulated tool sliding on a **Unitree Go1**, with no surface model.
 - Derived a friction-independent normal cue: out-of-plane drag vanishes exactly at the true surface normal.
 - Built the surface-frame estimator and a **100 Hz** centroidal MPC (**OCS2 SQP**) over a **1 kHz** WBC QP.
 - Validated on **hardware**: completed **5/5** wall and **5/5** column trials at **10 N**, with **2.29°/2.48°** normal error and **2.97/2.06 N** force RMSE.
 - Fixed-normal and friction-compensated baselines completed **0/5** column orbits (**p=0.008**, Fisher exact).
- **GaitNet: RL Footstep Planning** ALMaS Research Group, WPI
Graduate Researcher; Advisor: Dr. Mahdi Agheli Nov 2025 – May 2026
 - Co-first-authored **GaitNet**, combining learned foothold costs with RL gait selection over leg, displacement, and swing duration, conditioned on robot state, velocity commands, and terrain heightmaps.
 - Integrated GaitNet with **OCS2 perceptive MPC** and GPU elevation mapping; achieved **69.4%** survival on missing-step terrain in Isaac Lab versus **25.6%** for a single-leg baseline.
- **TOTO: Autonomous Mobile Robot** NIT Trichy, India
Undergraduate Researcher; Advisor: Dr. D. Ezhilarasi Sep 2024 – May 2025
 - Built **TOTO**, an autonomous mobile robot for real-time mapping and navigation on uneven terrain, using **ICP-SLAM** and an **Iterated Kalman Filter** to fuse encoder and RPLiDAR data.
 - Achieved $\pm 2\%$ velocity-tracking error on uneven terrain and reduced LiDAR point-cloud processing by **10%** using min–max filtering.
- **Bio-mimetic Autonomous Underwater Vehicle (BAUV)** IIT Madras, India
Research Intern; Advisor: Dr. Asokan Thondiyath Mar 2023 – Sep 2023
 - Developed sensing and motor-control hardware for a tuna-inspired biomimetic AUV, including a Jetson Orin interface PCB and **CAN** control for Maxon BLDC propulsion drives.
 - Supported Simulink-based tuning of the biomimetic propulsion controller.

Awards, Nominations and Fellowships

- **Dr. Glenn Yee Graduate Student Tuition Award** WPI, Worcester
Awarded to graduate students for academic excellence Fall 2026
- **Winner, Smart India Hackathon** Ministry of Education, India
Awarded in a nationwide student innovation competition 2023

Selected Projects

Code and demonstration videos: github.com/souravselvaraj

- **Simulated Ice-Skating Locomotion for Humanoids via Blade-Contact RL** [video] Jul 2026 – Present
 - Trained a **Unitree G1** to skate with PPO on passive 3mm blades without imitation data.
 - Skating gait emerges purely from anisotropic blade-ice contact.
 - Designed an asymmetric actor-critic, curriculum, slip-aware rewards, and dynamics randomization.
 - Tracked 0.9 m/s within **0.09 m/s** at **94% single support** with **2–4 cm/s** lateral slip.
- **Simulated Bimanual Lifting for Humanoids via Residual AMP-PPO** [code] May 2026 – Jul 2026
 - Trained a **Unitree G1** to lift objects with residual AMP-PPO over a frozen locomotion actor.
 - Held objects within **3 cm** of target using adaptive curricula and dynamics randomization.
- **Acrobatic Quadrotor Control with L1 Adaptive Augmentation** [code] Oct 2025 – Dec 2025
 - Built an **SE(3)** controller with **L1 adaptive** augmentation for a **Crazyflie 2.1** in ROS 2.
 - Achieved **2.2 cm** position RMSE and rejected **94%** of force-step disturbances.

Additional projects: Conditional Flow Matching visuomotor policies (matched or beat BC-RNN on 4/5 tasks, code); ORB-SLAM rover; visuomotor imitation (code); PDDL + OMPL planning.

Teaching Experience

- **RBE 521: Legged Robotics** Worcester Polytechnic Institute
Teaching Assistant & Laboratory Curriculum Developer; Instructor: Dr. Mahdi Agheli May 2026 – Aug 2026
 - Developed hands-on lab modules using Unitree Go1 and G1 robots.

Academic Service & Leadership

- **Reviewer:** *IEEE Transactions on Automation Science and Engineering (T-ASE)*, Jan–Feb 2026.
- **Team Lead, ISRO Robotics Challenge:** Led 10 members building an autonomous rover at NIT Trichy, Aug 2024–May 2025.

Technical Skills

Software: ROS 2, MuJoCo, Isaac Lab, Gazebo, PyTorch, OCS2, Pinocchio, Docker

Programming: C/C++, Python, MATLAB/Simulink

Robotics: MPC, Whole-Body Control, Trajectory Optimization, Reinforcement Learning, Imitation Learning

Hardware: Unitree Go1/G1, Franka Panda, Crazyflie 2.1, Jetson Orin, RealSense, F/T Sensors, CAN